ISA 444: Business Forecasting

10 - Nonseasonal Smoothing (Cont.)

Fadel M. Megahed

Associate Professor
Department of Information Systems and Analytics
Farmer School of Business
Miami University

fmegahed
@FadelMegahed
fmegahed@miamioh.edu
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Spring 2022

Outline

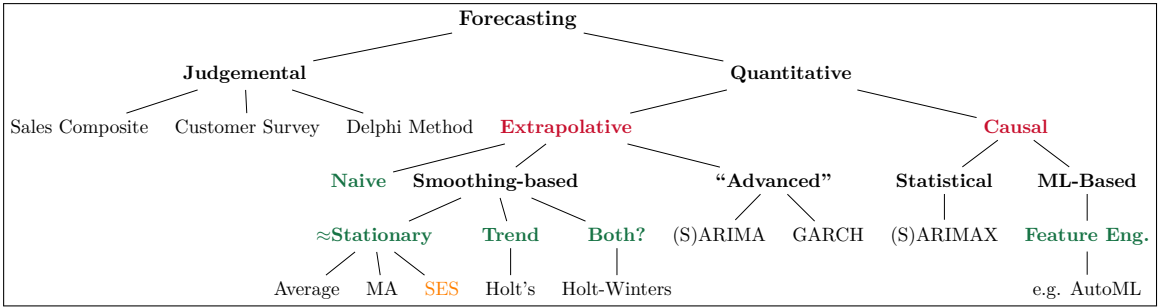
1. **Preface**
2. Exam 01 Review and Discussion
3. Simple Exponential Smoothing (SES)
4. Recap

What we Covered Last Class

Main Learning Outcomes

- ☑ **Describe and apply smoothing/forecasting with a cumulative average.**
- ☑ **Describe and apply forecasting with a moving average.**

Recap: A 10,000 Foot View of Forecasting Methods



A 10,000 foot view of forecasting techniques[1]

[1] An (incomplete) classification of forecasting techniques. Note that these focus on univariate time-series. Hence, they exclude popular approaches used in multivariate time series forecasting.

Learning Objectives for Today's Class

Main Learning Outcomes

- **Review and understand the concepts covered in Exam 01.**
- **Recognize time series that are appropriate for simple exponential smoothing (SES).**
- **Use SES to smooth past observations of a time series.**
- **Use SES to forecast future observations of a time series.**
- **Compare the forecasting performance of SES to other suitable techniques (i.e., methods that require similar assumptions).**

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2 Exam 01 Review and Discussion

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4 Recap

A Walk Through of Every Question on Exam 01

We will go through every question on Exam 01. **Please do not hesitate to ask me questions as we are going through this.**

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Definition and Basic Principles

Simple Exponential Smoothing (SES) is a method used for one-step-ahead forecasting of a time series when there is no trend or seasonal pattern, but the mean may drift slowly over time. The mean is said to have a "local level."

Similar to the idea behind a moving average, SES averages the values of the times series, but applies weights of decreasing importance to values that are farther away from the forecast. The weights of the observations "exponentially decay" as we move away from them in time.

The SES one-step-ahead forecast is given by:

$$l_{t+1} = l_t + \alpha(y_t - l_t) = \alpha y_t + (1 - \alpha) l_t, \tag{1}$$

where $0 < \alpha < 1$ is the smoothing parameter, and l_t is the level of the series at time t. Note that l_{1+1} is often denoted as f_{t+1} since it represents our one step-ahead forecast for $t + 1$.

Impact of the Smoothing Parameter: The Math

Let us examine what goes into the computations for l_4:

$$
\begin{aligned}
l_4 &= \alpha y_3 + (1-\alpha) l_3 \\
&= \alpha y_3 + (1-\alpha)[\alpha y_2 + (1-\alpha) l_2] \\
&= \alpha y_3 + \alpha(1-\alpha) y_2 + (1-\alpha)^2 l_2 \\
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\end{aligned}
\tag{2}
$$

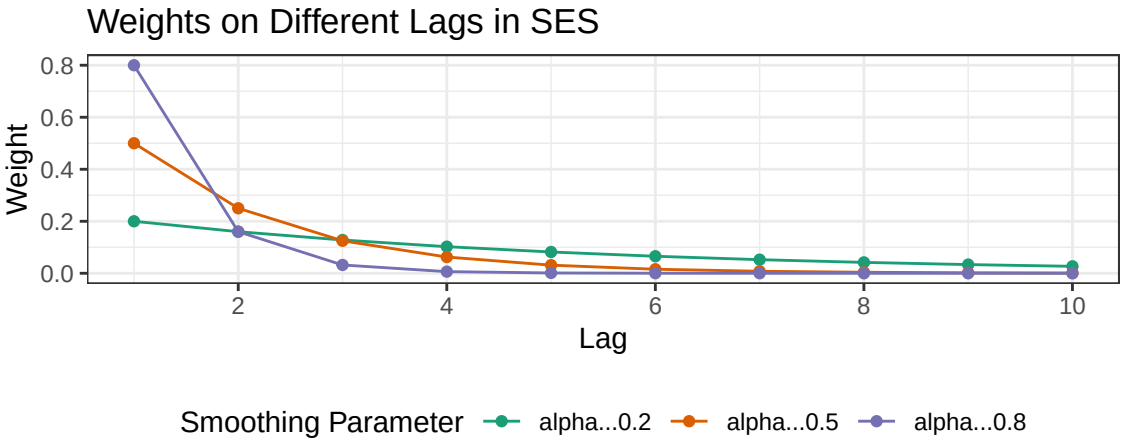
Note that SES needs two parameters: (a) the smoothing paramater α, and (b) the initial value for the level (i.e., l_0). Note that we will use $l_0 = l_1 = y_1$.

Impact of the Smoothing Parameter: The Math (Cont.)

For l_{10}, the weights of the observed values at t are distributed as follows:

t	alpha...0.2	alpha...0.5	alpha...0.8
9	0.20000	0.50000	0.80000
8	0.16000	0.25000	0.16000
7	0.12800	0.12500	0.03200
6	0.10240	0.06250	0.00640
5	0.08192	0.03125	0.00128
4	0.06554	0.01562	0.00026
3	0.05243	0.00781	0.00005
2	0.04194	0.00391	0.00001
1	0.03355	0.00195	0.00000
0	0.02684	0.00098	0.00000

Impact of the Smoothing Parameter: Visually



Example 1: Problem Definition

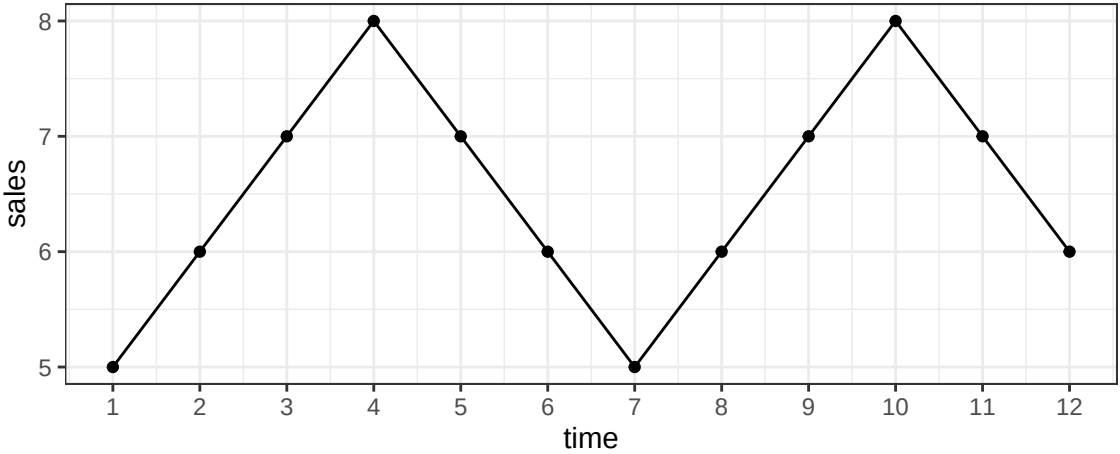
In this example, we are using R to forecast sales mimicking Table 3.3 in our reference book. See P. 72. Note the difference between the generated output here and that in the Table. If you wanted to exactly replicate that result, what minor edit to the R forecast is needed?

```
if(require(pacman)==FALSE) install.packages("pacman") # install pacman if not
pacman::p_load(magrittr, tidyverse, fpp2, scales) # load (and install if need

df = data.frame(time = 1:12,
                sales = c(5, 6, 7, 8, 7, 6, 5, 6, 7, 8, 7, 6))

df %>%
  ggplot(aes(x= time, y = sales)) + # setting the canvas
  geom_line() + geom_point() + # lines with dots highlighted
  scale_x_continuous(breaks = pretty_breaks(12)) + # making x_axis pretty (fr
```

Example 1: Charting the Original Data

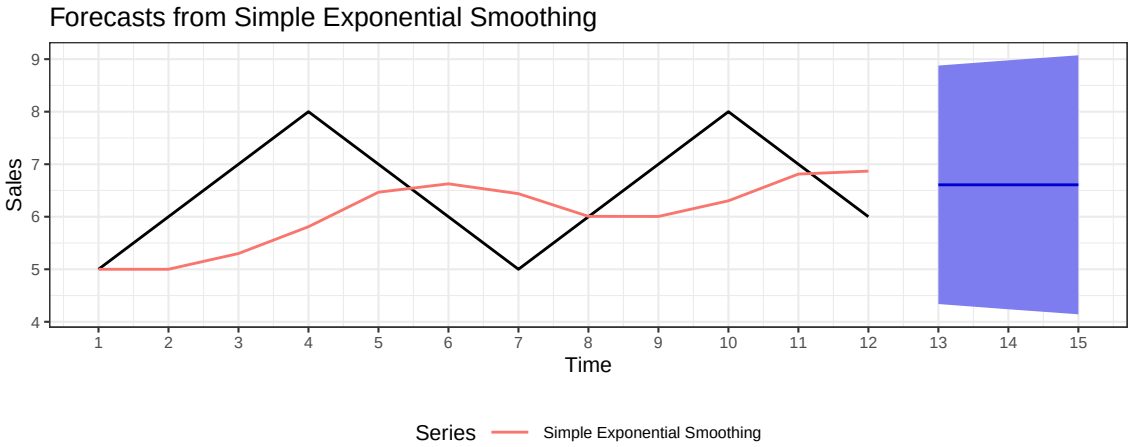


Example 1: Using R to Compute the SES

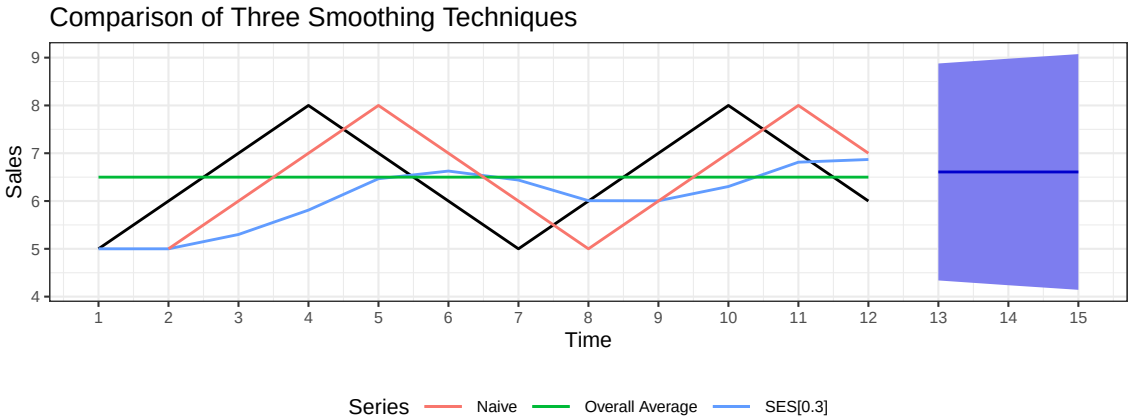
```
sales_ts = ts(df$sales, start = 1, frequency = 1)
sesRes = ses(sales_ts, initial = "simple", alpha = 0.3, h =3, level = 95)
summary(sesRes)
```

```
##
## Forecast method: Simple exponential smoothing
##
## Model Information:
## Simple exponential smoothing
##
## Call:
##  ses(y = sales_ts, h = 3, level = 95, initial = "simple", alpha = 0.3)
##
##   Smoothing parameters:
##     alpha = 0.3
```

Example 1: Using R to Chart the Data, SES & Forecast



Example 1: Comparing SES with Other Smoothing Methods



The blue ribbon/region captures the 95% PI based on the SES Forecast

Discussion Question

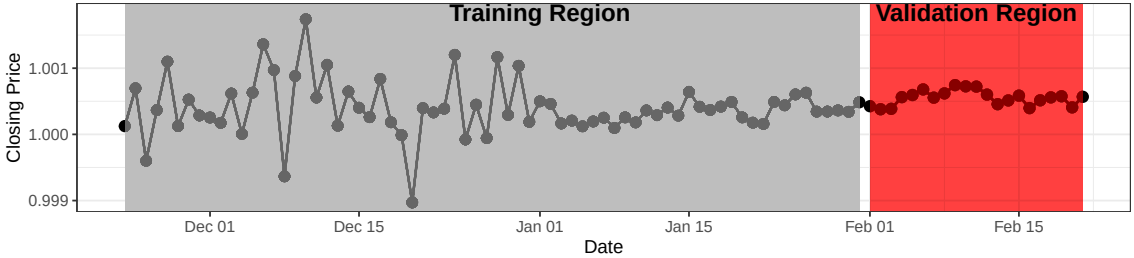
If you had to make a subjective choice for the value of the smoothing constant, what value would you choose for:

a. **a product with long-term steady sales and**
b. **a stock/cryptocurrency (e.g., AAPL, AMZN, BTC-USD, ADA-USD)?**

Training and Validation Samples [1]

Often you determine your smoothing parameter based on a training or baseline sample of observations, not the entire series. Then you apply the model using the smoothing parameter to the new observations and evaluate the fit on the out-of-sample observations.

Depicting Training (~77%) and Validation (~23%) Regions for the USDT−USD Data



Data from 2021−11−23 to 2022−02−21

Training and Validation Samples [2]

1. Determine the size of the training, or baseline sample.[2]
   a. Training sample size is usually 70-80% of the total available data.
   b. Training sample should maintain time order. With time series, the training sample usually consists of observations at the beginning of the sample, while validation sample consists of observations at the end of the available data.
2. Select the smoothing parameter based on the observations in the training sample only.
3. Evaluate the "in-sample" performance of the forecast using RMSE and graphs using the training sample.
4. Apply the model chosen in #2 to the validation sample.
5. Evaluate the "out-of-sample" performance of the forecast using RMSE and graphs.

[2] Slide is based on Dr. Allison Jones-Farmer's Handouts for ISA 444, Spring 2020.

Optimizing the Smoothing Parameter: WFJ Sales Series

To illustrate the aforementioned concepts, let us examine the data for the WFJ Sales Example (i.e., Example 3.2 in our reference book). Per the reference book example, we will use the first the 26 observations as the estimation sample.

```
pacman::p_load(readxl)
WFJ = read_excel("Data/WFJ_sales.xlsx") %>% select(c(1,2))
```

This example will be coded live in class to obtain the results below.

The optimal alpha obtained using R is equal to 0.727.

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	587.398	2970.310	2058.263	1.521	6.423	0.862	-0.199

The Validation Results

A continuation of the live coding session, where we: (a) examine the validation results; and (b) print the combined training and validation results in one table.

	ME	RMSE	MAE	MPE	MAPE
Training Set	587.398	2970.310	2058.263	1.521	6.423
Validation Set	-76.797	3915.053	2562.870	-0.931	7.336

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Summary of Main Points

Main Learning Outcomes

- **Recognize time series that are appropriate for simple exponential smoothing (SES).**
- **Use SES to smooth past observations of a time series.**
- **Use SES to forecast future observations of a time series.**
- **Compare the forecasting performance of SES to other suitable techniques (i.e., methods that require similar assumptions).**

Things to Do

- **Recommended:** Thoroughly read Chapter 3.1-3.3 of our reference book.
- Go through the slides, examples and make sure you have a good understanding of what we have covered.
- **Required:** Complete the graded assignment (see details in next slide).

Graded Assignment 07: Evaluating your Understanding

Please go to Canvas (click here) and answer the questions. **Due February 28, 2022 [8:30 AM]**

What/Why/Prep? The purpose of this assignment is to evaluate your understanding and retention of the material covered up to the end of Class 10. To reinforce your understanding of the covered material, I also suggest reading Chapter 3.1-3.3 of the reference book.

General Guidelines:

- Individual assignment.
- This is **NOT** a timed assignment.
- Proctorio is NOT required for this assignment.
- You will need to have R installed (or accessible through the Remote Desktop)

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